Mahdi Alam

(647)-916-4783 | mahdi.alam@ryerson.ca | https://www.linkedin.com/in/mahdi-alam-264127150/ |
https://github.com/mahdi-alam360

Research Focus

Currently conducting MSc research on improving self-supervised deep learning via curriculum learning and adaptive optimization, with emphasis on reproducible experimentation, quantitative evaluation, and scalable training pipelines.

Education

Toronto Metropolitan (Ryerson) University September 2023 – May 2025
MSc, in Artificial Intelligence via Research-based Thesis *Toronto, ON*

Toronto Metropolitan (Ryerson) University September 2019 – May 2023
BSc. (Hons), in Computer Science *Toronto, ON*

Relevant Coursework

Graduate: Machine Learning, Computer Vision, Research Methods in Computer Science
Undergraduate: Machine Learning, Algorithms, Probability & Statistics, Artificial Intelligence

Technical Skills

Programming: Python, C++, C
Machine Learning: Deep Learning, Self-Supervised Learning, Computer Vision, Optimization, Transformers
Libraries/Frameworks: PyTorch, TensorFlow, NumPy, SciPy, Pandas, OpenCV, Scikit-learn
Research Systems: Experiment Design, Ablation Studies, Reproducibility, Git, Docker, Linux

Research Experience

Machine Learning Researcher (Medical Image Registration) Aug 2019 – Present
Toronto Metropolitan University (TMU) *Toronto, ON*

- Designed a two-phase coarse-to-fine training strategy combining multi-resolution curriculum learning and adaptive optimization for self-supervised deep learning.
- Extended the On-the-Fly Guidance (OFG) framework without modifying model architecture or inference pathways.
- Improved Dice Similarity Coefficient (DSC) by up to +1.7 points across public MRI and CT datasets (LPBA40, IXI, OASIS).
- Evaluated deformation quality via Jacobian determinant analysis and conducted reproducible experiments with ablations and benchmark evaluation in Python-based deep learning pipelines.

Publications and Manuscripts

- **Mahdi Alam**. *Enhancing On-the-Fly Guidance for Medical Image Registration through Multi-Resolution Curriculum and Adaptive Optimization.* Manuscript in preparation, 2025.

Projects

Search Engine (NLP / Retrieval) | *Python* Oct 2022 – Nov 2022

- Implemented TF–IDF and BM25 retrieval; trained Naïve Bayes and linear SVM baselines; compared retrieval effectiveness across models.

Colour Clustering Pipeline | *Python, Scikit-learn* Sep 2023 – Dec 2023

- Extracted dominant colour distributions and applied K-means clustering to identify recurring palettes and evaluate clustering stability.